Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 7 - solutions

October 30, 2026

Exercise 1. The Selection Sort algorithm sorts the input sequence $a_1, ..., a_n$ of real numbers in ascending order

Algorithm 1 Selection Sort

```
1:  for i = 1 to n − 1 do
2:      min ← i + 1
3:      for j = i + 1 to n do
4:          if a_min > a_j then
5:              min ← j
6:          end if
7:      end for
8:      if a_i > a_min then
9:          swap a_i and a_min
10:     end if
11: end for
```

Which sequence will result in the most operations when using Selection Sort (swaps and comparisons)?

◯ 9, 3, 8, 6, 5, 1, 7

✓ 4, 7, 2, 9, 5, 6, 3

◯ 0, 1, 2, 3, 4, 5, 6

◯ 7, 6, 5, 4, 3, 2, 1

Number of comparisons and swaps for selection sort on input: (9, 3, 8, 6, 5, 1, 7)

Iteration	nb comparisons on line 4	nb comparisons on line 8	nb swaps	Total nb operations
i=1	6	1	1	8
i=2	5	1	0	14
i=3	4	1	1	20
i=4	3	1	0	24
i=5	2	1	1	28
i=6	1	1	1	31

Number of comparisons and swaps for selection sort on input: (4, 7, 2, 9, 5, 6, 3)

Iteration	nb comparisons on line 4	nb comparisons on line 8	nb swaps	Total nb operations
i=1	6	1	1	8
i=2	5	1	1	15
i=3	4	1	0	20
i=4	3	1	1	25
i=5	2	1	1	29
i=6	1	1	1	32

Number of comparisons and swaps for selection sort on input: (0, 1, 2, 3, 4, 5, 6)

Iteration	nb comparisons on line 4	nb comparisons on line 8	nb swaps	Total nb operations
i=1	6	1	0	7
i=2	5	1	0	13
i=3	4	1	0	18
i=4	3	1	0	22
i=5	2	1	0	25
i=6	1	1	0	27

Number of comparisons and swaps for selection sort on input: (7, 6, 5, 4, 3, 2, 1)

Iteration	nb comparisons on line 4	nb comparisons on line 8	nb swaps	Total nb operations
i=1	6	1	1	8
i=2	5	1	1	15
i=3	4	1	1	21
i=4	3	1	0	25
i=5	2	1	0	28
i=6	1	1	0	30

Exercise 2.

$$L_{x_1} = (y_3, y_1, y_2) \qquad L_{y_1} = (x_2, x_1, x_3)$$
$$L_{x_2} = (y_2, y_3, y_1) \qquad L_{y_2} = (x_1, x_3, x_2)$$
$$L_{x_3} = (y_1, y_2, y_3) \qquad L_{y_3} = (x_3, x_2, x_1)$$

Let L_x for $x \in X = \{x_1, x_2, x_3\}$ be the preference list of x as given above and let L_y for $y \in Y = \{y_1, y_2, y_3\}$ be the preference list of y as given above.
We say that a matching is X-optimal (resp. Y-optimal) if all elements of X (resp. Y) are matched to its most-preferred partner among all partners it can have in any stable matching.
The matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$ is

◯ *unstable.*

◯ *stable and Y-optimal.*

◯ *stable and X-optimal.*

✓ *stable but not a stable matching that is X- or Y-optimal.*

In the following matching: $\{(x_1, y_3), (x_2, y_2), (x_3, y_1)\}$, all $x \in X$ are matched with their optimal choices and therefore the matching $\{(x_1, y_3), (x_2, y_2), (x_3, y_1)\}$ is stable and X-optimal. Because this matching $\{(x_1, y_3), (x_2, y_2), (x_3, y_1)\}$ is different from the given matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$, the matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$ is not X-optimal, so that the third answer is not correct. (Note that in the matching $\{(x_1, y_3), (x_2, y_2), (x_3, y_1)\}$ all $y \in Y$ are matched to their least favorite choices.)

In the following matching: $\{(x_1, y_2), (x_2, y_1), (x_3, y_3)\}$, all $y \in Y$ are matched with their optimal choices and therefore the matching $\{(x_1, y_2), (x_2, y_1), (x_3, y_3)\}$ is stable and Y-optimal. Again, because this matching $\{(x_1, y_2), (x_2, y_1), (x_3, y_3)\}$ is different from the given matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$, the matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$ is not Y-optimal, so that the second answer is not correct. (Note that in the matching $\{(x_1, y_2), (x_2, y_1), (x_3, y_3)\}$ all $x \in X$ are matched to their least favorite choices.)

Actually, in the matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$, no $x \in X$ or $y \in Y$ is matched with its optimal or with its worst choice. So, in principle, there could be quite a lot of room to move things around:

in (x_1, y_1):

x_1 would prefer to be matched with y_3, but y_3 prefers its current x_2 to x_1. Because there is no other $y \in Y$ that x_1 would prefer to y_1, the pair (x_1, y_1) is stable from x_1's point of view.

y_1 would prefer to be matched with x_2, but x_2 prefers its current y_3 to y_1. Because there is no other $x \in X$ that y_1 would prefer to x_1, the pair (x_1, y_1) is stable from y_1's point of view as well.

It follows that the pair (x_1, y_1) is stable.

in (x_2, y_3)**:**

x_2 would prefer to be matched with y_2, but y_2 prefers its current x_3 to x_2. Because there is no other $y \in Y$ that x_2 would prefer to y_3, the pair (x_2, y_3) is stable from x_2's point of view.

y_3 would prefer to be matched with x_3, but x_3 prefers its current y_2 to y_3. Because there is no other $x \in X$ that y_3 would prefer to x_2, the pair (x_2, y_3) is stable from y_3's point of view as well.

It follows that the pair (x_2, y_3) is stable.

in (x_3, y_2)**:** a similar argumentation can be used again, or we can simply argue that the pair (x_3, y_2) must be stable because its instability would imply instability of another pair which is impossible because all other pairs are stable.

It follows that only the fourth answer is correct.

Exercise 3. Use the cashier's algorithm to make change using quarters (25 cents), dimes (10 cents), nickels (5 cents), and pennies (1 cent) for:

1. 87 cents.
2. 49 cents.
3. 99 cents.
4. 33 cents.

Are the solutions found optimal?
Solution:
We apply the cashier algorithm in full detail only for the first case as it's pretty similar for other values of n.

1. Given total amount: n = 87
   (a) First pass (quarters) $c_1 = 25$
   We note that $n \geq 25$, thus we add 1 quarter and decrease the total amount by 25.

$$d_1 = 1$$

$$n = n_{old} - 25 = 87 - 25 = 62$$

   We note that $n \geq 25$, thus we add 1 quarter and decrease the total amount by 25.

$$d_1 = 1 + 1 = 2$$

$$n = n_{old} - 25 = 62 - 25 = 37$$

   We note that $n \geq 25$, thus we add 1 quarter and decrease the total amount by 25.

$$d_1 = 2 + 1 = 3$$

$$n = n_{old} - 25 = 37 - 25 = 12$$

   We note n < 25 thus the first pass ends.

(b) Second pass (dimes) $c_2 = 10$
We note that $n \geq 10$, thus we add 1 dime and decrease the total amount by 10.

$$d_1 = 3$$
$$d_2 = 1$$
$$n = n_{old} - 10 = 12 - 10 = 2$$

We note n < 10 thus the second pass ends.

(c) Third pass (nickels) $c_3 = 5$
We note n < 5 thus the third pass ends.

(d) Fourth pass (pennies) $c_4 = 1$
We note that $n \geq 1$, thus we add 1 penny and decrease the total amount by 1.

$$d_1 = 3$$
$$d_2 = 1$$
$$d_4 = 1$$
$$n = n_{old} - 1 = 2 - 1 = 1$$

We note that $n \geq 1$, thus we add 1 penny and decrease the total amount by 1.

$$d_1 = 3$$
$$d_2 = 1$$
$$d_4 = 2$$
$$n = n_{old} - 1 = 1 - 1 = 0$$

We note n < 1 thus the fourth pass ends.
Conclusion: 87 cents is 3 quarters, 1 dime, 0 nickels and 2 pennies.

2. (similarly) 49 cents is 1 quarter, 2 dimes, 0 nickels and 4 pennies.
3. (similarly) 99 cents is 3 quarters, 2 dimes, 0 nickels and 4 pennies.
4. (similarly) 33 cents is 1 quarter, 0 dimes, 1 nickel and 3 pennies.

Exercise 4. *Let $\{A, B, C, D\}$ be a set of men, and $\{a, b, c, d\}$ a set of women. We want to match up men and women using the Gale–Shapley algorithm in two different ways. The preferences of men and women are given in the following lists, going from most preferable on the left to least preferable on the right.*

Men	*1st*	*2nd*	*3rd*	*4th*
A	*c*	*d*	*b*	*a*
B	*d*	*c*	*a*	*b*
C	*a*	*c*	*b*	*d*
D	*b*	*d*	*a*	*c*

Women	*1st*	*2nd*	*3rd*	*4th*
a	*D*	*A*	*B*	*C*
b	*C*	*B*	*A*	*D*
c	*C*	*B*	*A*	*D*
d	*D*	*A*	*B*	*C*

1. *If the men propose, and women accept/reject, what is the matching after the algorithm terminates?*

   After all men propose to their 1st prefered woman, since they are all different, they accept and the algorithm terminates with the following matching:

   A-c, B-d, C-a, D-b.

2. *If the women propose, and men accept/reject, what is the matching after the algorithm terminates?*

   Firstly a proposes to D, and b proposes to C, and they accept because it is the first proposal.

   a-D, b-C; c,d,A,B unmatched

   Then c proposes to C. Since C prefers c more than b, b gets rejected, and C accepts c instead.

   a-D, c-C; b,d,A,B unmatched

   Now d proposes to D, and again because of preferences of D, a gets rejected and D accepts d instead.

   c-C, d-D; a,b,A,B unmatched

   a and b are now unmatched, so they proposed to their second preference, A and B respectively. Since they are free, A and B accept and the algorithm terminates with:

   a-A, b-B, c-C, d-D

3. *Who is the best possible (stable) valid partner for "a"?*

   First we will show that a matching where a is paired with D is unstable. This is because D prefers d to a, and d will prefer D whoever she is matched with (D is her 1st preference). So a cannot be matched with D. On the other side a can be matched with A, as we have seen from 2. So the best matching for a is A.

   One can also simply argue that the solution found in 2 above is Women-optimal, so no woman can do better in any stable matching than found under 2, so A is the best partner than a can find.

Exercise 5. *Which function below grows fastest when n goes to infinity?*

- ✓ $(\log_3(33))^{n-3}$
- ◯ 3^n
- ◯ $n^{3\log_3(n)}$
- ◯ $n^3 \log_3(n)$

$(\log_3(33))^{n-3}$ grows faster than 3^n:
Because $3 = \log_3(3^3) = \log_3(27) < \log_3(33)$ it follows that

$$\frac{(\log_3(33))^n}{3^n} = \Big(\frac{\log_3(33)}{3}\Big)^n$$

goes to infinity for $n \to \infty$. Because the constant factor $(\log_3(33))^{-3}$ does not affect the growth rate, it follows that $(\log_3(33))^{n-3}$ grows faster than 3^n.
3^n grows faster than $n^{3\log_3(n)}$:
Let us note that $\forall x > 0$, we have $x = 3^{\log_3(x)}$. By replacing x by $n^{3\log_3(n)}$, we get:
$n^{3\log_3(n)} = 3^{\log_3(n^{3\log_3(n)})} = 3^{3\log_3(n)\log_3(n)} = 3^{3\log_3(n)^2}$.
Because n grows faster than $3\log_3(n)^2$, it follows that 3^n grows faster than $3^{3\log_3(n)^2} = n^{3\log_3(n)}$
$n^{3\log_3(n)}$ grows faster than $n^3 \log_3(n)$:
$n^{3\log_3(n)}$ grows faster than $n^3 \log_3(n)$ because the $\log_3(n)$ is in the exponent and is greater than 1 for all $n > 3$

Exercise 6. Suppose that $f_1(x)$ is $O(g_1(x))$ and that $f_2(x)$ is $O(g_2(x))$.
Show that $(f_1 + f_2)(x)$ is $O(\max(|g_1(x)|, |g_2(x)|))$:
Solution:
Let us define g as the max of absolute values of g_1 and g_2 : $g(x) = \max(|g_1(x)|, |g_2(x)|)$
Since $f_1(x)$ is $O(g_1(x))$ and $f_2(x)$ is $O(g_2(x))$, there are constants C_1, C_2, k_1 and k_2 such that:

$$|f_1(x)| \leq C_1|g_1(x)| \quad \forall x > k_1$$
$$|f_2(x)| \leq C_2|g_2(x)| \quad \forall x > k_2$$

To majorize the sum of $f_1(x)$ and $f_2(x)$, note that

$$\begin{aligned}
|(f_1 + f_2)(x)| &= |f_1(x) + f_2(x)| \\
&\leq |f_1(x)| + |f_2(x)| \\
&\leq C_1|g_1(x)| + C_2|g_2(x)| \ \ \forall x > \max(k_1, k_2) \\
&\leq C_1|g(x)| + C_2|g(x)| \\
&= (C_1 + C_2)|g(x)|
\end{aligned}$$

Consequently, by setting $C = C_1 + C_2$ and $k = \max(k_1, k_2)$, we can conclude that $f(x)$ is $O(\max(|g_1(x)|, |g_2(x)|))$.

Exercise 7. [**Exam 2023**] Order the following functions so that each function is big-O of the next function. For example if f is $O(g)$ and g is $O(h)$, the order should be f, g, h. Justify your choices.

$$\begin{aligned}
f_1(n) &= 2^{100000} \\
f_2(n) &= \left(\log_2(n^2)\right)^2 \\
f_3(n) &= 0.001n! \\
f_4(n) &= n^{0.01}
\end{aligned}$$

f_1 is constant whereas all other functions are increasing towards infinity when n tends to infinity and therefore, f_1 must be the first element of the list.
We know that $(\log_b(n))^c$ is $O(n^d)$ for $b > 1$ and $c, d > 0$. Hence, f_2 is $O(f_4)$.
Finally, by using the 3 properties below and the fact that if $f(x)$ is $O(g(x))$ and $g(x)$ is $O(h(x))$, then $f(x)$ is $O(h(x))$, we can conclude that f_4 is $O(f_3)$.

- n^d is $O(b^n)$ for $d > 0$, $b > 1$
- b^n is $O(c^n)$ for $c > b > 1$
- c^n is $O(n!)$ for $c > 1$

We have the following ordering: f_1, f_2, f_4, f_3.